

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Aman Varshney
Program/Batch	B082-Networkwalks
Date	22 August 2026
Modules completed	W2-PM4 (The Harvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using The Harvester Tool (W2-PM4) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
theHarvester	To gather the list of email IDs related to NETWORKWALKS
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using theHarvester tool inside Kali linux Virtual Machine.

First, I used theHarvester command in Kali linux terminal to initiate the tool.

I then used “\$ theHarvester -d networkwalks.com -l 1000 -b baidu” command to find the IPs, emails, & hosts related to NETWORKWALKS.

Then I used “\$ theHarvester -d networkwalks.com -l 50 -b all” command to search all the sources to provide me 50 email IDs & sub-domains related to NETWORKWALKS.

Finally, I took screenshots of the outputs and also copied the outputs in a separate text document to mention in the report and Github repository.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows ipconfig command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified one live host:

- 192.168.XX.X

The example results also included one MAC address.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Publicly available organizational information	theHarvester was used against networkwalks.com using multiple search sources to identify publicly available information associated with the organization.	Publicly exposed information may assist an attacker during reconnaissance and help in building an initial profile of the organization.	● Medium
2	Domain-related information identifiable	theHarvester was executed with -d networkwalks.com, allowing publicly indexed domain-related information to be collected.	Domain information can contribute to further reconnaissance and identification of the organization's external attack surface.	● Low
3	Search-engine reconnaissance possible	theHarvester command using -b baidu was used to collect information indexed by the Baidu search engine.	Search-engine-indexed information may reveal publicly accessible organizational details that could support further enumeration.	● Low
4	Multi-source reconnaissance possible	theHarvester was also executed with -b all to gather information from its available supported sources.	Information collected from multiple sources can be correlated to create a broader profile of the organization's publicly exposed footprint.	● Medium
5	One live host identified on local network	Zenmap ping scan was performed against the user's own IPv4 address. Only one live host was detected,	The scan confirms the presence of the tested host on the local network. Since no other live hosts were identified, no additional	● Low

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
		corresponding to the user's own device.	exposed LAN hosts were observed during this scan.	

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or email addresses does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Minimize Public Information Exposure

Regularly review publicly available organizational information and remove unnecessary sensitive details.

2. Monitor the External Attack Surface

Conduct periodic assessments to identify newly exposed domains, services, and organizational information.

3. Review DNS and Web Configuration

Regularly check DNS records, HTTP responses, and web configurations to minimize unnecessary technical information disclosure.

4. Perform Regular Network Scanning

Authorized network scans should be conducted periodically to identify unexpected or unauthorized hosts and services.

5. Keep Systems and Software Updated

Web applications, plugins, operating systems, and other software components should be regularly updated and patched to address known security vulnerabilities.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities related to **footprinting, reconnaissance, and network scanning**.

In the footprinting activity, I used **theHarvester** to gather publicly available information related to the **networkwalks.com** domain. I performed searches using different sources and learned how reconnaissance can help in collecting useful information about an organization's digital presence.

In the network scanning activity, I used **Zenmap** to perform a ping scan on my own IPv4 address and local network. The scan identified **one live host, which was my own device**, helping me understand how active hosts can be identified within a network.

These exercises helped me understand the importance of **information gathering and network reconnaissance in cybersecurity**. I also gained practical experience with theHarvester and Zenmap and learned how these tools can be used in an authorized security assessment to understand an organization's publicly exposed information and network environment.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

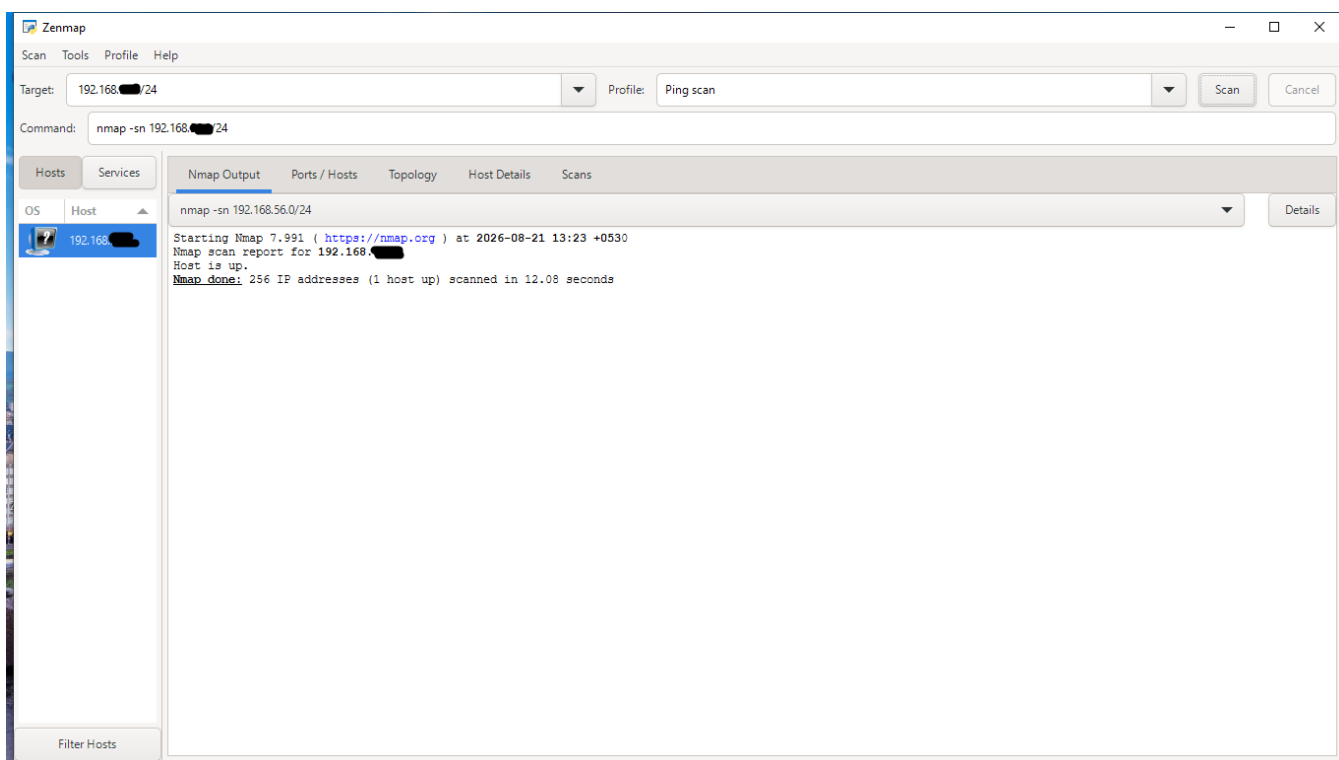
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

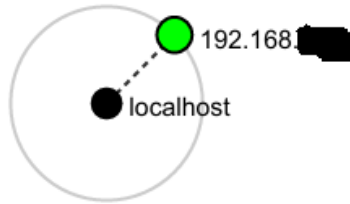
8. Evidences Collected

```
kali@kali: ~  
Session Actions Edit View Help  
$ theHarvester -h  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
* theHarvester 4.11.1  
* Coded by Christian Martorella  
* Edge-Security Research  
* cmartorella@edge-security.com  
*  
*****  
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME]  
               [-w WORDLIST] [-a] [-q] [-b SOURCE]  
  
theHarvester is used to gather open source intelligence (OSINT) on a company or domain.  
  
options:  
-h, --help                show this help message and exit  
-d, --domain DOMAIN        Company name or domain to search.  
-l, --limit LIMIT          Limit the number of search results, default=500.  
-s, --start START          Start with result number X, default=0.  
-p, --proxies              Use proxies for requests, enter proxies in proxies.yaml.  
-s, --shodan               Use Shodan to query discovered hosts.  
--screenshot SCREENSHOT    Take screenshots of resolved domains specify output directory: --screenshot output_directory  
-e, --dns-server DNS_SERVER DNS server to use for lookup.  
-t, --take-over            Check for takeovers.  
-r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.  
-n, --dns-lookup           Enable DNS server lookup, default False.  
-c, --dns-brute            Perform a DNS brute force on the domain.  
-f, --filename FILENAME
```

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ theHarvester -d networkwalks.com -l 1000 -b baidu  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
* theHarvester 4.11.1  
* Coded by Christian Martorella  
* Edge-Security Research  
* cmartorella@edge-security.com  
*  
*****  
[*] Target: networkwalks.com  
[*] Searching Baidu.  
[*] No IPs found.  
[*] No emails found.  
[*] No people found.  
[*] No hosts found.
```

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ theHarvester -d networkwalks.com -l 50 -b all  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
* [ASCII Art] *  
* theHarvester 4.11.1 *  
* Coded by Christian Martorella *  
* Edge-Security Research *  
* cmartorella@edge-security.com *  
*****  
[*] Target: networkwalks.com  
  
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml  
Failed to process bevigil search for word: 'networkwalks.com'  
Error Message:  
[!] Missing API key for bevigil.  
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml  
  
[!] Missing API key for Bitbucket.  
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml  
Failed to process bufferoverun search for word: 'networkwalks.com'  
Error Message:  
[!] Missing API key for bufferoverun.
```





-End-

Author

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub